Leonardo Salicari, Ph.D.

Curriculum Vitae

Bologna, Italy
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Working Experience

Apr 2025 – ongoing **HPC Scientific Software Engineer**, *CINECA*, Italy
Working in the Scientific Computing and Application Performance team, Life Science area. Working in the EU-India collaboration GANANA (Workflow developer and responsable of the ML immunologic pipeline project), EPICURE support team and teaching programming and HPC topics.

Oct 2021 – Dic 2024 **Ph.D. in Physics**, *University of Padua*, Italy
Supervisor: Prof. Antonio Trovato

Apr 2024 – May 2024 **Teaching assistant**, *University of Padua*, Italy
Laboratory assistant for engineering students in the course of Physics Laboratory I (Mechanics)

2019 – 2023 **Tutoring**, *Cooperativa L'Impronta/independently*, Italy
Tutoring highschool students in: Mathemathics, Physics and Programming (C/C++)

June 2015 – July 2019 **Children's chaperone**, *Colleverde Summer Camps*, Colle Umberto (PG, Italy)

Education

Oct 2021 – Dic 2024 **Ph.D. in Physics**, *University of Padua*, Excellent
Supervisor: Prof. Antonio Trovato
Thesis's title: Statistical mechanics and numerical simulations of topologically entangled proteins
Statistical Mechanics group website

Oct 2018 – Apr 2021 **Master's degree in Physics - Statistical Mechanics**, *University of Padua*, *110/110 with honors*
Selected topics: Stochastic dynamics, Machine Learning (supervised), Data Analysis and Visualization, Non-Equilibrium Statistical Mechanics, Molecular Dynamics, and Biophysics

Oct 2015 – Sept 2018 **Bachelor's degree in Physics**, *University of Perugia*, *110/110 with honours*

Skills

Languages Python, C/C++, Cython, LaTeX, bash
Strong proficiency acquired through University courses, Ph.D., and activities in CINECA. Example of a recent developed project: immuno-workflow. Proficient with agentic tools for code developing.

HPC WMS, profiling tools
Worked with pre-Exa scale HPC systems to profile and optimize applications. Developed workflows for large-scale MD simulations.

DevOps git, CI/CD tools, containerization, github/lab
Software versioning skills and CI tools knowledge developed for personal and research projects.

Data An. Polars/Pandas, Scipy, Matplotlib, Numpy
Example project on entangled membrane proteins

ML PyTorch, Keras, Scikit-learn
Example project: a CNN + NN model to classify knots in polymers. Expertise in using these libraries in HPC

Teaching Tutoring and teaching assistant experience. Here a repository with materials created.

Achievements

Public. Fields: Computational Biophysics, Numerical Simulations, Statistical Mechanics, Complex Systems
Salicari, Leonardo; Baiesi, Marco; Orlandini, Enzo; Trovato, Antonio: Folding kinetics of an entangled protein. In: *PLOS Computational Biology* 19 (2023), 11, Nr. 11, 1–30. https://doi.org/10.1371/journal.pcbi.1011107
Salicari, Leonardo; Trovato, Antonio: Entangled Motifs in Membrane Protein Structures. *International Journal of Molecular Sciences* 24 (2023), Nr. 11, 9193. https://www.mdpi.com/1422-0067/24/11/9193

Talks Three invited talks and two accepted ones in international conferences during the Ph.D. period (Example: Invited webinar for BioExcel)

Awards Best poster at BioExcel School on Biomolecular simulations (Apr 2022)

Selected coursework

HPC CUDA/OpenACC, OpenMP and MPI, Building systems, nsys.

Ph.D. HPC (CUDA, OpenMP, MPI, TBB), Molecular Dynamics, QM methods, Statistical Physics of Deep Learning, Teaching, Containerization.

Master Stochastic dynamics, Machine Learning (classical and deep learning), Data Analysis and Visualization, Non-Equilibrium Statistical Mechanics, Molecular Dynamics, and Biophysics.

Bachelor Advance Calculus, Linear algebra, Statistics, Computational Physics, OOP (C++) and Computer Architecture.

Languages

Italian Native speaker

English C1 *Academically and professionally, I wrote various dissertations and held public talks in English. Based on the CEFR levels. Completed the "Academic English Course" (30h) at the UNIPD Language Center*

Soft Skills

Planning and Management *from Ph.D., tutoring experiences, and GANANA project management*

Problem Solving *Research experience in STEM field*

Teamwork *from tutoring, research and EU projects*

Other

Driving licence: B

Basic of cardiopulmonary resuscitation (CPR): Open Badge